Cambridge International Examinations
Cambridge International General Certificate of Secondary Education

CANDIDATE NAME	
CENTRE NUMBER	
CANDIDATE NUMBER	

BIOLOGY **0610/41**

Paper 4 Theory (Extended) **October/November 2016**

1 hour 15 minutes

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

Electronic calculators may be used.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

The syllabus is approved for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

This document consists of **18** printed pages and **2** blank pages.

DC (CW/FD) 117738/4
© UCLES 2016

CAMBRIDGE International Examinations

www.root19.com +91 - 9969 353 391

BLANK PAGE

© UCLES 2016
www.root19.com

+91 - 9969 353 391

1 Penicillin is an antibiotic.

(a) **(i)** Explain why doctors give antibiotics to people who are ill.

.....

.....

.....

.....[2]

(ii) Explain why it is important to complete a full treatment of antibiotics.

.....

.....

.....

.....

.....

.....

.....[3]

© UCLES 2016

www.root19.com
+91 - 9969 353 301

(b) Penicillin was discovered in 1928 by Alexander Fleming.

Name the type of microorganism that produces the antibiotic penicillin.

.....[1]

(c) Penicillin is produced commercially in fermenters as shown in Fig. 1.1.

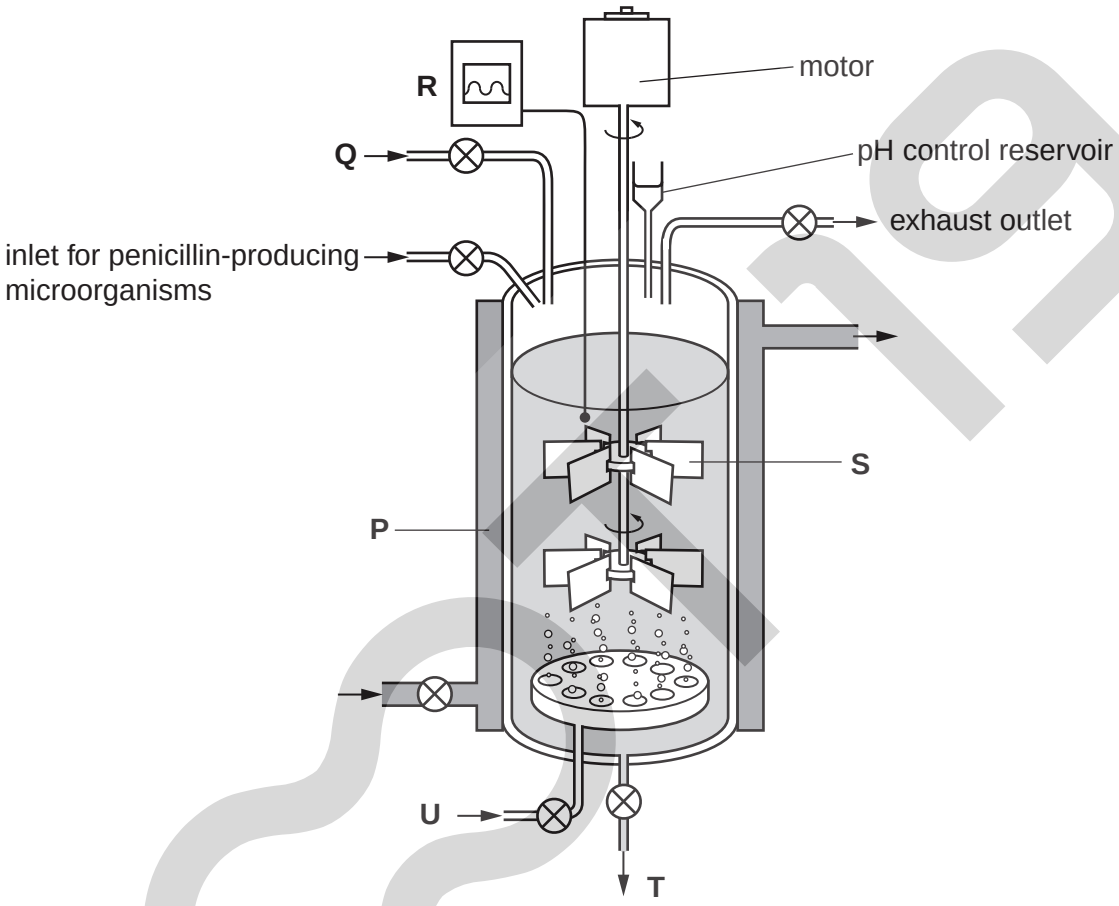

Fig. 1.1

www.root19.com +91 - 9969 353 391

(i) Describe how a fermenter can be sterilised.

.....

.....

.....[2]

(ii) Table 1.1 shows some names of the parts of the fermenter and their functions.

Complete Table 1.1.

One row has been done for you.

Table 1.1

letter from Fig. 1.1	name	function
	water jacket	
S		
	nutrient inlet	
R		
	air supply	
T	outlet	allows collection of the liquid containing penicillin after fermentation

[5]

(d) Describe what happens to the liquid containing penicillin after it is collected from the fermenter.

.....

.....[1]

[Total: 14]

© UCLES 2016

www.root19.com
+91 - 9969 353 301

2 Carp are a type of fish. Researchers in Brazil measured the body lengths of a population of carp in a river in **1998** and again in **2008**.

Histograms of their results are shown in Fig. 2.1.

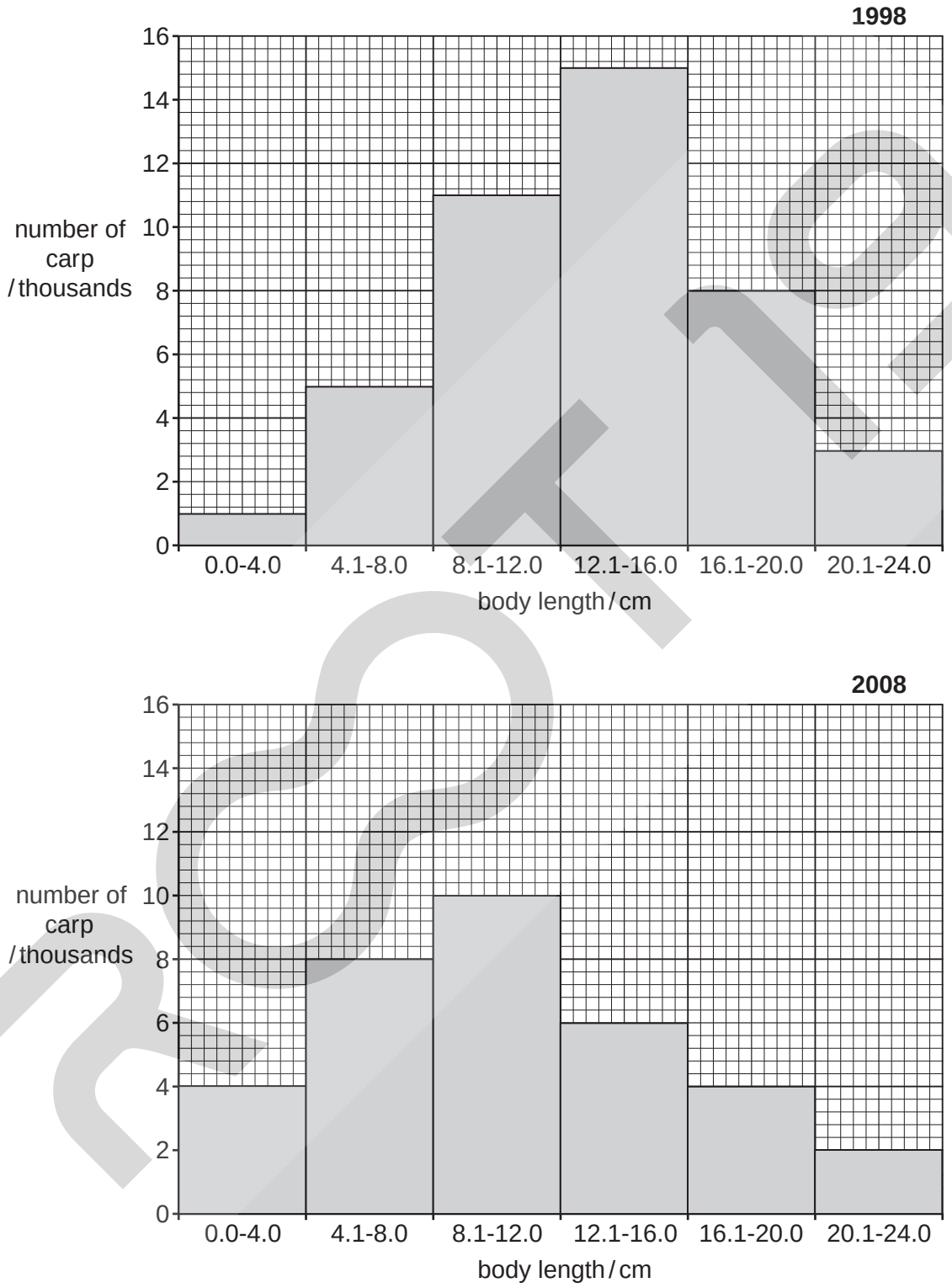

Fig. 2.1

www.root19.com +91 - 9969 353 391

(a) Define the term *population*.

.....

.....

.....

.....

.....[2]

(b) Describe the variation in body length of the carp population in **1998**. Use the data in Fig. 2.1 to support your answer.

.....

.....

.....

.....

.....

.....

.....[3]

(c) The total population of carp in 1998 was 43 000 fish.

(i) Calculate the total population of carp in **2008**.

Show your working.

.....[2]

root19

www.root19.com +91 - 9969 353 301

(ii) The decrease in the carp population by 2008 was caused by overfishing.

Explain how fish stocks can be sustained.

.....

.....

.....

.....

.....

.....

.....[4]

(d) Body length is an example of continuous variation.

(i) Suggest what causes the variation in body length in a population of fish.

.....

.....

.....

.....

.....

.....

.....[2]

(ii) Continuous variation is shown with a histogram.

Name the type of graph that should be used to show **discontinuous** variation.

.....[1]

[Total: 14]

© UCLES 2016
www.root19.com

+91 - 9969 353 391

3 Pepsin is a protease enzyme found in the alimentary canal.

(a) **(i)** Name the product formed from the digestion of proteins by protease enzymes.

.....[1]

(ii) State the organ in the alimentary canal where pepsin is secreted.

.....[1]

(b) A biologist performed an experiment to find the optimum pH for the activity of pepsin.

The enzyme activity was measured in four test-tubes. Each test-tube contained a 1 cm^3 cube of cooked egg white which contains protein.

Fig. 3.1 shows the four test-tubes.

The biologist measured the time taken for the complete digestion of the cubes of cooked egg white.

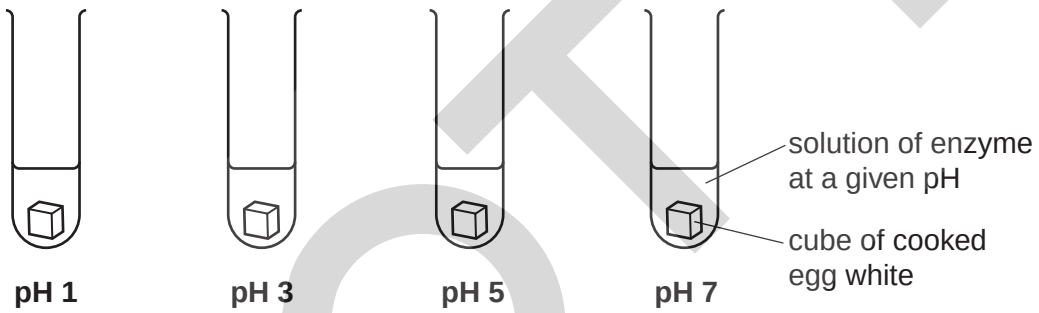

Fig. 3.1

(i) The biologist ensured all the cubes of cooked egg white were exactly the same size.

Suggest why.

.....

.....

.....

.....

.....[2]

(ii) Temperature must be controlled in this experiment.

Describe how temperature could be controlled.

.....

.....[2]

www.root19.com +91 - 9969 353 301

(c) The same experiment was performed with trypsin, another protease enzyme.

Predict the optimum pH for trypsin.

.....[1]

(d) Enzymes, such as proteases, are important in digestion.

Describe in detail how enzymes function, using other digestive enzymes as examples.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[6]

[Total: 13]

© UCLES 2016
www.root19.com

+91 - 9969 353 391

4 Type 1 diabetes is caused by the immune system destroying body cells.

(a) **(i)** Suggest which organ in the body is attacked by the immune system to cause Type 1 diabetes.

.....[1]

(ii) Antibodies are part of the immune system.

Describe how antibodies function.

.....

.....

.....

.....

.....[2]

(b) Humans need vitamin D as part of their diet.

(i) Describe a cause of vitamin D deficiency in humans.

.....

.....[1]

(ii) Describe the effects of vitamin D deficiency in humans.

.....

.....

.....

.....

.....[2]

root 19

www.root19.com +91 - 9969 353 301

Scientists have found that mice can suffer from Type 1 diabetes. They also found that vitamin D affects the development of Type 1 diabetes in mice.

Two groups of mice were studied. One group were normal and the other group were vitamin D-deficient. The percentage of mice in each group that developed Type 1 diabetes was recorded every fifty days.

Table 4.1 shows their results.

Table 4.1

age / days	percentage of mice with Type 1 diabetes	
	normal mice	vitamin D-deficient mice
0	0	0
50	1	1
100	8	35
150	34	62
200	45	65
250	46	65

(c) Describe the effect of vitamin D deficiency on the development of Type 1 diabetes in mice. Use the data in Table 4.1 to support your answer.

.....

.....

.....

.....

.....

.....

.....[3]

(d) Suggest the symptoms that mice with Type 1 diabetes would have.

.....

.....

.....

.....

.....

.....

.....[4]

www.root19.com +91 - 9969 353 391

(e) Describe the treatment for human patients with Type 1 diabetes.

.....

.....

.....

.....

.....

.....

.....

.....

.....[3]

[Total: 16]

© UCLES 2016

www.root19.com
+91 - 9969 353 301

5 Fig. 5.1 is an image of a germinating seed showing the growing root.

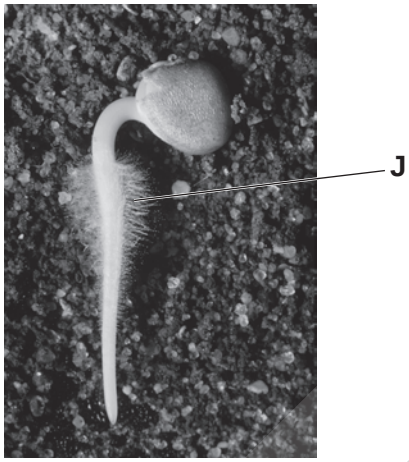

Fig. 5.1

(a) Describe **and** explain how the structures seen at **J** are adapted for their function.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[5]

ROOT 19

www.root19.com +91 - 9969 353 391

(b) **(i)** The root shown in Fig. 5.1 is growing downward into the soil.

Name this response seen in roots.

.....[1]

(ii) Name the chemical that controls this response.

.....[1]

(iii) There are situations, either in wild plants or in laboratory experiments, where roots do not grow downwards.

Suggest and explain **one** situation.

.....

.....

.....

.....

.....[2]

[Total: 9]

ROOT 19

www.root19.com +91 - 9969 353 301

BLANK PAGE

© UCLES 2016
www.root19.com

+91 - 9969 353 391

6 A DNA molecule has two strands as shown in Fig. 6.1.

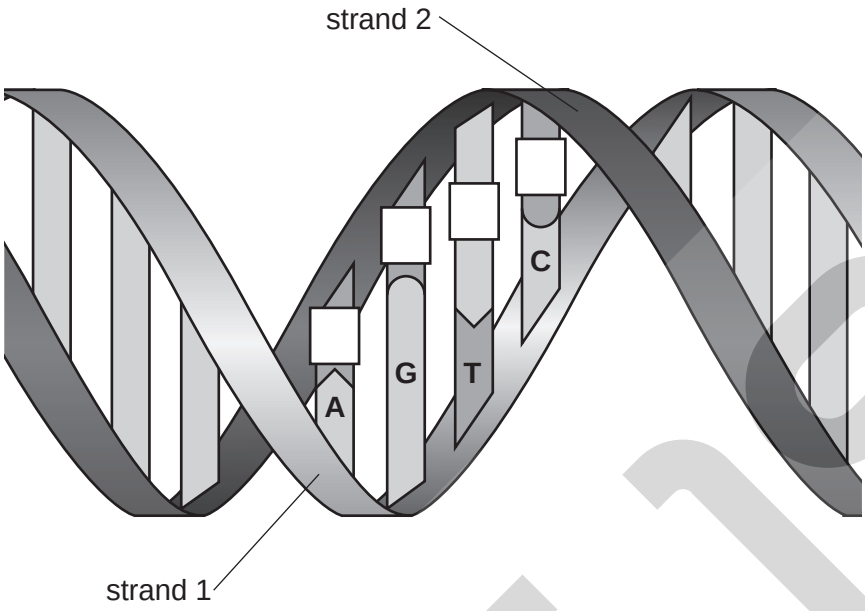

Fig. 6.1

(a) **(i)** Fill in the boxes on Fig. 6.1 to show the letter of the bases on strand 2 that will pair with the corresponding bases on strand 1. [2]

(ii) State the name for the structure of a DNA molecule as shown in Fig. 6.1.

.....[1]

www.root19.com +91 - 9969 353 301

When molecules of DNA are used to classify species, only one of the two DNA strands is sequenced.

First the DNA sequence from one strand of a DNA molecule from each species is lined up against one strand from another species.

The bases of the DNA sequences from the same strand can then be compared with each other.

Fig. 6.2 shows a short section from the DNA sequences of eight plant species. There are ten differences between species **A** and species **B**. These differences are shown in Fig. 6.2.

```
Species A:  CTCCTCGGGT GACGGCCTAG CCCGTTGACG AATCCCATTC CTAAACTTTT
Species B:  CTCCTAGGGT GCAGGACTAG CCCGTTGACG AATCCCATTC CCAAGA
Species C:  CTCATAGGGT GCAGGCCTAG CCCGTTGACG AATCACATTC CGATT
Species D:  CTCATAGGGT GCAGGCCTAG CCCCTTGACG AATCCAATTC CGCTT
Species E:  CTCATAGGGT GCAGGCCTAG CCCGTTGACG AATCCAATTC CGCTT
Species F:  CTCCTAGGTT GCAGGCCTAG CCCTTTGAAG AATCACATTC CCCAA
Species G:  CTCCTCGGGT GCAGGCATAG CCCTTTGACG AATCCCCTTC CGAAA
Species H:  CTCCTAGGGT GCAGGCATAG CCCTTTGACG AATCCCCTTC CAAAAT
```

Fig. 6.2

(b) The number of differences between the DNA sequences of the eight species shown in Fig. 6.2 are recorded in Table 6.1.

Count the number of differences between the DNA sequences shown in Fig. 6.2 for:

- species **C** and species **D**
- species **G** and species **H**

Write your answers in Table 6.1. [2]

www.root19.com +91 - 9969 353 391

Table 6.1

	species **A**	species **B**	species **C**	species **D**	species **E**	species **F**	species **G**	species **H**
species **A**		10	10	13	12	11	10	9
species **B**			7	8	7	7	7	6
species **C**					3	7	8	8
species **D**					1	9	9	8
species **E**						9	8	10
species **F**							6	7
species **G**								
species **H**								

(c) The most closely related species have the fewest differences between their DNA sequences.

State which **two** plant species shown in Table 6.1 are most **distantly related** to each other.

.....[1]

(d) The most closely related species have the shortest distance from a branching point on a classification tree.

Use the information in Table 6.1 to complete the classification tree in Fig. 6.3. Write the letter corresponding to species **B**, **C**, **D** and **G** in the box next to the correct branch of the classification tree. [3]

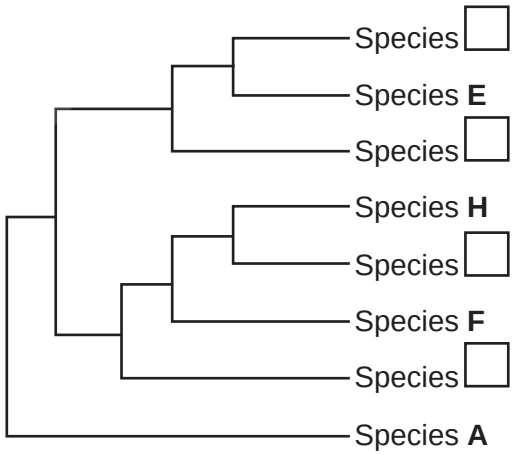

Fig. 6.3

www.root19.com +91 - 9969 353 301

(e) A modern method for improving crop productivity is to cut out sections of DNA carrying a useful gene from one organism and place them into another organism.

(i) Name the technique of inserting genes from one organism into another.

.....[1]

(ii) A gene for producing a vaccine has been inserted into banana plants.

Give **two** other examples in which crop plants have been changed by inserting genes. State one advantage for each example.

example 1 .....

advantage .....

.....

example 2 .....

advantage .....

.....
[4]

[Total: 14]

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge International Examinations Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cie.org.uk after the live examination series.

Cambridge International Examinations is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of University of Cambridge Local Examinations Syndicate (UCLES), which is itself a department of the University of Cambridge.

www.root19.com +91 - 9969 353 391